

Is-A RELATIONSHIP :

- The relationship between two objects which is similar to the parent and child relation is known as the Is-A relationship. In Java IS-A relationship is represented by Inheritance.

INHERITANCE : The process of one class acquiring all the properties and behaviour from the other class is called inheritance. Inheritance is an OOP feature in Java that providing code reusability, method overriding, and an IS-A relationship between classes.

Parent class => A class becomes a parent class when it serves as a blueprint or template from which other classes inherit properties and behaviours. A parent class, also known as a Superclass. It contains common attributes and methods that are shared by its subclasses.

Child class => A child class, also known as a subclass, is a class that inherits properties and behaviours (methods) from another class. The child class can extend the functionality of the parent class.

Java uses the **extends** keyword to create inheritance.

```
class Parent {  
  
}  
  
class Child extends Parent {  
  
}
```

Note: Here, A child class can reuse the fields and methods of the parent class. This reduces code duplication.

Basic Points to Remember During Inheritance in Java

***Inheritance in Java is unidirectional:** Unidirectional inheritance means that the relationship flows only in one direction from parent class to child class. The child class inherits the properties and methods of the parent class. The parent class cannot inherit or access the specific properties and methods of the child class.

***Constructors are not inherited:** A child class cannot inherit the constructor of the parent class. However, the parent constructor is called automatically when a child object is created.

***Private members are not accessible in the child class:** private variables and methods belong only to the parent class.

Revise App Code:

1. Shape App
2. Fruit Market
3. Banking